

Sanika Govardhan Sangvai

AI/ML Engineer

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LinkedIn | GitHub | Portfolio | LeetCode

Education

VIT Bhopal University

B.Tech, Computer Science and Engineering (Health Informatics)

2023 – 2027

CGPA: 8.62/10

Skills

- **Languages:** C++, SQL
- **Frontend:** HTML5, CSS3
- **Databases:** MySQL
- **AI/ML & Data:** Machine Learning, Deep Learning, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Computer Vision, YOLOv8, OpenCV
- **Tools & Platforms:** GitHub, vscode

Projects

Skin Disease Detection using CNN

Feb 2025

Python, TensorFlow, Keras, CNN, Computer Vision

GitHub

- Developed a CNN-based image classification model using TensorFlow/Keras to detect 7 categories of skin diseases, including melanoma, eczema, and psoriasis, from dermoscopic images
- Trained and validated the model on the HAM10000 dataset, applying data augmentation techniques to improve generalization across lesion types
- Addressed severe class imbalance in the dataset using targeted resampling and augmentation strategies, improving minority-class recognition
- Achieved 90%+ test accuracy through iterative architecture tuning and hyperparameter experimentation

Smart Agriculture System – AI-Based Irrigation Prediction

May 2026

Python, Scikit-learn, FastAPI, Streamlit, ESP8266, ThingSpeak

GitHub

- Built an IoT-to-cloud pipeline using ESP8266 sensors (soil moisture, temperature, humidity) streaming real-time data to ThingSpeak
- Trained and compared Decision Tree, Random Forest, and XGBoost classifiers for irrigation prediction, selecting Random Forest for its stability on noisy sensor data
- Developed a FastAPI backend to serve model predictions and a Streamlit dashboard visualizing live sensor readings and next-irrigation timing
- Designed the end-to-end architecture (sensors → cloud → ML inference → dashboard) as a fully automated decision pipeline

RetailPulse – Real-Time Customer Footfall Analytics

Nov 2025

Python, YOLOv8, OpenCV, FastAPI, SQLite

GitHub

- Developed a real-time computer vision pipeline using YOLOv8 for person detection and persistent multi-object tracking to prevent double-counting
- Implemented directional entry/exit counting using vector cross-product logic to compute live store occupancy
- Automated timestamped event storage enabling historical footfall analysis beyond live monitoring

Achievements

- **Top 25 Performer**, NeoCodeathon 2026 – Recognized for coding and problem-solving performance
- **Rank 29/250+**, AMD Ryzen Slingshot Regional Ideathon – VIT Bhopal University
- **Health Hack 2024** (VIT Bhopal – JHU) – Built an IoT-based Remote Health Monitoring System

Certifications

- Applied Machine Learning in Python – Coursera, Feb 2026
- Data Analysis with Python – IBM SkillsBuild, Sep 2025
- AWS Certification Course – Intellipaat, Sep 2025